

WEEK 1 TASK

Designing a Scalable IoT Architecture

Week 1 Technical Design & Simulation Report

Yash Ingle

B.Tech – Electronics & Communication Engineering (Advanced Communication Technology)

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1. Abstract

This report presents a practical, scalable IoT architecture for industrial and environmental monitoring. The proposed design combines sensor/edge devices, local gateways, IP networking, MQTT messaging, cloud storage, analytics and dashboards. A hybrid topology is used so that short-range devices can communicate efficiently with gateways while long-range or low-power sensors can use LoRaWAN where appropriate. Edge processing filters and aggregates data before cloud transmission, reducing bandwidth usage and improving response time. The architecture is designed to expand from approximately 10 devices to 100, 1,000 and beyond by adding gateways and horizontally scaling cloud services.

2. Objectives

- Design an IoT architecture that supports increasing numbers of sensors and devices.
- Define device interconnections, network topology, protocols and end-to-end data flow.
- Use edge computing to reduce unnecessary cloud traffic and provide fast local decisions.
- Identify scalability, reliability, security and operational risks and provide mitigations.
- Define a simulation plan using Cisco Packet Tracer or GNS3 and measurable performance targets.
- Document assumptions and design choices for practical deployment.

3. Proposed Use Case and Assumptions

Use case: a multi-zone smart industrial monitoring environment in which temperature, humidity, gas, vibration and motion sensors report equipment and environmental conditions. Each zone contains sensor nodes connected to an edge gateway. Gateways forward normalized telemetry to an MQTT broker and cloud services. Alerts requiring immediate action can be generated at the edge, while historical analytics are handled in the cloud.

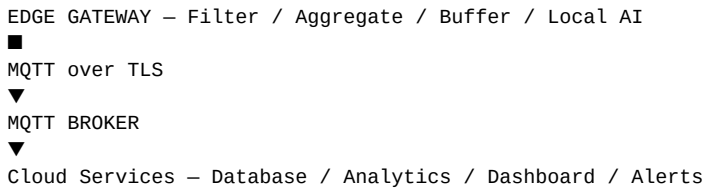
- Initial deployment: 10–100 sensor/device nodes.
- Growth objective: 1,000+ devices by adding gateways and cloud capacity.
- Sensor messages are small telemetry packets.
- MQTT is primary telemetry; HTTPS/REST is used for web/API integration.
- TLS, authentication and device identity are mandatory in production.
- Simulation values are design targets until measured in the selected simulator.

4. IoT Architecture

Layer	Main components	Primary function
Device	ESP32/MCU, temperature, humidity, gas, vibration, motion sensors	Sense, locally preprocess and transmit telemetry
Edge/Gateway	Raspberry Pi/industrial IoT gateway, local broker/cache	Protocol translation, filtering, aggregation, buffering and local alerts
Network	Wi-Fi, Ethernet, LoRaWAN, IP routing	Reliable and scalable connectivity
Cloud/Application	MQTT broker, database, analytics, dashboard, alert service	Storage, visualization, analytics, device management

Structural layout

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SENSORS / DEVICES
■■■ ESP32 + Temperature/Humidity
■■■ Gas / Smoke
■■■ Vibration
■■■ Motion / Other Sensors
■
Wi-Fi / LoRaWAN / Ethernet
▼
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5. Network Topology and Protocol Selection

Technology/Protocol	Role	Reason for selection
I ² C / SPI / UART	Sensor-to-MCU	Simple, low-cost local interfaces
WiFi	Local wireless IP connectivity	High availability and easy integration in buildings
Ethernet	Gateway/backhaul	Stable, high-throughput wired connection
LoRaWAN	Long-range low-power sensors	Useful for remote/low-data-rate devices
MQTT	Telemetry messaging	Lightweight publish/subscribe model and scalable topic hierarchy
HTTPS/REST	Application/API integration	Standard secure web communication
TLS	Transport security	Encrypts telemetry and protects credentials
DNS/DHCP	IP network services	Automated addressing and service discovery

6. Data Flow and MQTT Topic Design

Sensor → MCU → Gateway → Validation/Filtering → MQTT Broker → Cloud Ingestion → Database → Analytics → Dashboard/Alert

- factory/zone01/device001/temperature
- factory/zone01/device001/humidity
- factory/zone02/device015/vibration
- factory/zone02/device015/status

7. Scalability Design

Scale	Architecture response
10–100 devices	One or more gateways, centralized MQTT broker, basic dashboard
100–1,000 devices	Multiple gateways by zone, topic-based routing, gateway buffering and cloud horizontal scaling
1,000–10,000+ devices	Gateway clusters, broker clustering, load balancing, partitioned/time-series storage and automated device provisioning

- Edge filtering
- Horizontal scaling
- Automated device provisioning
- Gateway buffering
- Load balancing
- Database partitioning and retention
- Observability
- Signed OTA firmware updates

8. Simulation Plan – Cisco Packet Tracer / GNS3

- Scenario A – 10 devices: verify basic connectivity and routing.
- Scenario B – 50 devices: measure increased traffic and gateway utilization.
- Scenario C – 100 devices: test congestion, packet loss and latency.
- Optional Scenario D – 500+ logical devices: evaluate architectural scaling limits.

Record packet delivery ratio, average latency, packet loss, link utilization, gateway CPU/memory (if supported), and application response time. Insert topology and traffic screenshots after running the simulation.

9. Expected Performance Metrics

Metric	Design target	Purpose
Packet delivery ratio	>95%	Measures reliability under normal load
Local edge response	<100 ms	Supports rapid local alerts/control
Cloud dashboard update	<1–2 s	Provides near-real-time monitoring
Gateway availability	>99%	Reduces single-point failure risk
Scalability	10× device growth	Demonstrates expansion without redesign
Security	TLS + authentication	Protects telemetry and device identity

These are engineering targets, not claimed measured results. Replace them with actual simulator observations before final submission.

10. Security Architecture

- Unique device identities; no shared default credentials.
- TLS for MQTT and API traffic.
- Role-based access control.
- VLAN/firewall segmentation.
- Disable unused services and ports.
- Signed OTA firmware and rollback.
- Security logging and monitoring.
- Key/certificate rotation and revocation.

11. Risk Assessment and Mitigation

Network congestion	High	Edge filtering, QoS, gateway segmentation and capacity planning
Gateway failure	High	Redundant gateways, watchdogs, local buffering and failover
Cyber attack	Very High	TLS, authentication, firewalling, network segmentation and monitoring
Sensor failure	Medium	Heartbeat/status messages, calibration schedule and replacement strategy
Cloud/WAN outage	High	Local buffering and edge alerts; synchronize when connection returns
Rapid device growth	High	Modular gateways, MQTT clustering and horizontal cloud scaling
Power failure	High	UPS/battery backup for gateways and critical infrastructure
Bad/duplicate data	Medium	Schema validation, timestamps, sequence IDs and deduplication

12. Reliability and Fault-Tolerance

The gateway is an important failure boundary. Sensor nodes can continue local sensing if the WAN is unavailable. A gateway queue stores unsent telemetry and forwards it after reconnection. Larger deployments should use overlapping coverage or standby gateways, while cloud services should be replicated and monitored.

13. Implementation Roadmap

Phase 1	Architecture & requirements	Define sensors, message rates, coverage, security and KPIs
Phase 2	Prototype	Build 2–5 sensor nodes and one gateway
Phase 3	Network simulation	Model 10/50/100 device loads and record metrics
Phase 4	Cloud integration	Deploy MQTT, database, dashboard and alerting
Phase 5	Security hardening	TLS, identities, segmentation, logging and OTA
Phase 6	Scale test	Add gateways/devices and verify performance targets

14. Design Justification

The hybrid edge-cloud approach is selected because IoT workloads are distributed and often generate frequent, small telemetry messages. Edge filtering reduces bandwidth use and WAN dependency while providing fast local response. MQTT decouples producers from consumers through publish/subscribe messaging, allowing multiple applications to consume the same telemetry stream.

15. Future Enhancements

- AI/ML anomaly detection.
- Digital twins.
- 5G/private cellular connectivity.
- Containerized services.
- Automated certificate provisioning.
- Predictive maintenance.

16. Conclusion

The proposed scalable IoT architecture combines heterogeneous sensors, edge gateways, IP networking, MQTT messaging and cloud applications into a modular system. The hierarchical hybrid topology avoids a flat network, while edge processing, buffering, security controls and horizontal scaling provide a practical path from prototype to large deployment. The next step is to run the three simulation scenarios and replace design targets with measured results.

17. Simulation Evidence Checklist

- Topology screenshot.
- Connectivity/packet test results.
- 10-device traffic screenshot.
- 50-device traffic screenshot.
- 100-device traffic screenshot.
- Measured latency/packet-loss/bandwidth table.
- Scaling comparison.

18. References / Suggested Technical Sources

- MQTT.org — MQTT specifications and overview.

- Cisco Networking Academy / Cisco Packet Tracer — networking and IoT simulation resources.
- LoRa Alliance — LoRaWAN technical resources.
- IETF — TLS and Internet standards.
- ISO/IEC 27001 family — information security management concepts.